remodelling. Osteoporosis is most common among post-menopausal women.

Negative calcium balance, which amounts to a 25% loss of bone mass between ages is 40 and 80, can be due to

1. low calcium intake
2. decreased intestine absorption
3. decreased reabsorption in the kidneys
4. decreased production of active metabolite of vitamin D normally formed in the kidneys, low protein or phosphorus intake, high fibre intakes or low oestrogen levels.
5. Vertebral fractures are also common in people with osteoporosis.
6. The compact bone mass in the vertebrate is responsible for the height loss, lower back pain, and humped backs characteristics of many elderly people.
7. Dissolution of the jawbone is another frequent symptom of osteoporosis; it is some major contributing factor in periodontal disease resulting in premature loss of teeth.

c) Neurological dysfunction

Problems of disorientation and slowing of neurological functioning, both seen in the elderly, have been attributed to various nutrient factors. A lack of niacin has long been associated with the dementia and depression of pellagra; a deficiency of chlorine hampers the synthesis of neurotransmitter acetylcholine; deficiencies of Vitamin B_6 and thiamine are associated with central nervous system problems.

d) Anaemia

Anaemia, characterized by fatigue, anxiety, lack of energy and sleeplessness, is a common result of inadequate iron. Iron inadequacy can be caused by low dietary intake; impaired absorption possibly resulting from lack of heme iron or Vitamin C or blood loss.

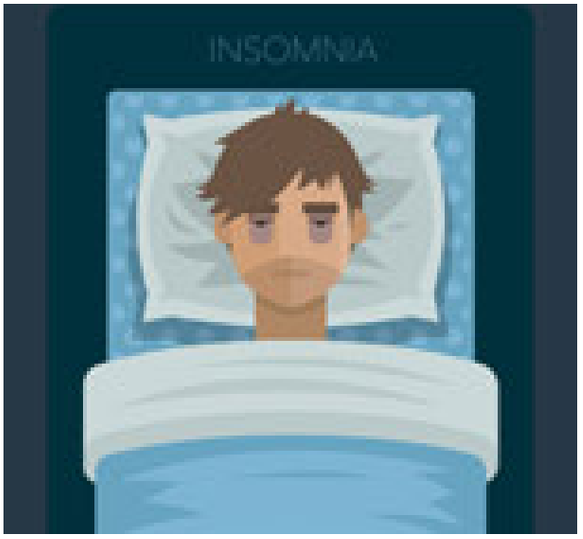


Fig 4.4 Insomnia

e) Drug related malnutrition

Usage of many of these influence the intake of nutrients through changes in taste, smell or unpleasant side effects, particularly in the gastrointestinal tract, which influences appetite and food intake. They also affect absorption of nutrients by binding the nutrients in the intestinal tract or by causing changes in the intestinal mucosa that reduces its capacity to accept nutrients for absorption. Because the rate of drug metabolism and detoxification in the liver is much slower in older people, drugs remain in the body longer to exit their influences on their metabolism of nutrients.

f) Food induced malnutrition

The extensive use of certain foods as implication for nutritional adequacy. The use of large amounts of tea increases the need for thiamine. The high phytate content of oatmeal and the tannin in tea reduce absorption of both iron and zinc. The use of foods high in added phosphorus creates and undesirably low calcium/phosphorus ratio that leads to poor bone calcification, especially when calcium intake is low. Extensive use of alcohol not only leads to diets of reduced nutrient content and multiple deficiencies but also has adverse effects on nutritional absorption.

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